

BERKE TAYFUN AKSEKI

Software Engineer | Backend, Data and Platform

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PROFESSIONAL SUMMARY

Backend and data engineer, 2026 Management and Computer Science graduate of LUISS Guido Carli, Rome. I design REST APIs in FastAPI, Node.js and ASP.NET Core, model PostgreSQL schemas, write queue-driven asynchronous workers with idempotency and retry semantics, and build Airflow and dbt pipelines gated by data-quality tests. The management half of the degree shows in how I work: written specifications, recorded decisions, measurable acceptance criteria.

TECHNICAL SKILLS

Programming Python, TypeScript, JavaScript, C#, SQL, R

Backend and APIs FastAPI, Node.js, ASP.NET Core, Django, REST API design, SQLAlchemy, Pydantic, JWT/OAuth2, async workers, job queues, idempotency, retries, circuit breakers

Data and pipelines PostgreSQL, SQLite, schema design, window functions, CTEs, Airflow, PySpark, dbt, Parquet, ETL/ELT, data-quality testing

Cloud and DevOps Docker, Docker Compose, AWS (ECS/Fargate, RDS, CDK), GitHub Actions CI/CD, Prometheus, Grafana, structured logging, Git

Analysis and ML Pandas, NumPy, SciPy, statsmodels, scikit-learn, A/B testing, difference-in-differences, cohort and funnel analysis, Power BI

Management and delivery Requirements and scope definition, technical specifications, decision records, acceptance criteria, stakeholder coordination

PROJECTS

Energy Connect Platform | backend service

TypeScript, Node.js, PostgreSQL, AWS CDK, Docker, Prometheus, Grafana | github.com/berketayfunakseki/energy-connect-platform

- Built a REST API and asynchronous worker tier over a durable PostgreSQL command queue, using SELECT ... FOR UPDATE SKIP LOCKED so no job is ever claimed or executed twice.
- Made every state-changing endpoint idempotent under a client-supplied key, and put exponential-backoff retries behind a circuit breaker, so a retry never duplicates a write and a failing downstream degrades instead of cascading.
- Instrumented it with structured logging, Prometheus metrics and Grafana dashboards against explicit SLOs; provisioned ECS/Fargate, RDS and CloudWatch as code with AWS CDK.

Project PARCA | deterministic rules engine and game server

Aug 2026 to present | C# / .NET, ASP.NET Core, JavaScript, Blender | private repository, walkthrough on request

- Built a deterministic rules engine in C# for a multi-day, eight-player strategy game: a seeded RNG replays an entire season exactly, so any disputed result is reproducible and auditable.
- Wrote and maintain a 146-case regression suite over eight negotiation minigames, transfer rules, elimination boundaries and season-end conditions; currently 146 passed, 0 failed.
- Built the ASP.NET Core server on a server-authoritative UTC clock and a transfer ledger, so no client can alter a balance, a deadline or a result.
- Own the product side as well: design document, economy specification and a log of 14 recorded decisions, each tied to a testable hypothesis. 174 commits in three weeks.

Analytics Data Platform | batch ETL pipeline and warehouse

Python, Apache Airflow, PySpark, dbt, PostgreSQL, Docker Compose | github.com/berketayfunakseki/analytics-data-platform

- Built a batch ETL pipeline: raw CSV through PySpark validation and deduplication into Parquet and PostgreSQL, then dbt staging models and daily-sales marts.
- Gated the marts behind dbt data-quality tests, so a bad load fails the Airflow DAG instead of silently reaching a dashboard.
- Packaged Airflow, Spark, PostgreSQL and dbt in Docker Compose, so a reviewer reproduces every result locally with one command.

Also on GitHub: Jobscope (FastAPI job aggregator, TF-IDF ranking) | Regulatory RAG Assistant (FAISS semantic search, CI) | Causal Lab (A/B testing, difference-in-differences) | SQL Analytics Lab | Churn Prediction ML API | Oral-B Geo-Analytics, a client project in KNIME and Power BI.

EXPERIENCE

Creative Manager Intern | LUISS Language Cafe, Rome, Italy

Feb 2026 to May 2026

- Owned creative direction and end-to-end delivery of the programme's communications across the university's official channels.
- Coordinated stakeholders across university departments on message, format and timing, to fixed publication deadlines.
- Translated programme objectives into a concrete content plan and acted as the single point of contact for its audience.

EDUCATION

BSc Management and Computer Science (L-18) | LUISS Guido Carli University, Rome, Italy

Sep 2023 to Jul 2026

- Joint degree: algorithms, databases and big data, AI and machine learning, statistics and quantitative models, plus the management and economics core.
- Thesis: macroeconomic resilience of 21 Italian regions, a 2015-2024 panel in R with difference-in-differences, two-way fixed effects and an event study.

LANGUAGES AND WORK AUTHORISATION

Languages: Turkish (native) | English (C1) | French (B2) | Italian (B1)

Work authorisation: Turkish citizen, no work permit required in Turkey. For Italy, entitled to a 12-month post-graduation residence permit convertible to a work permit, no employer quota wait.